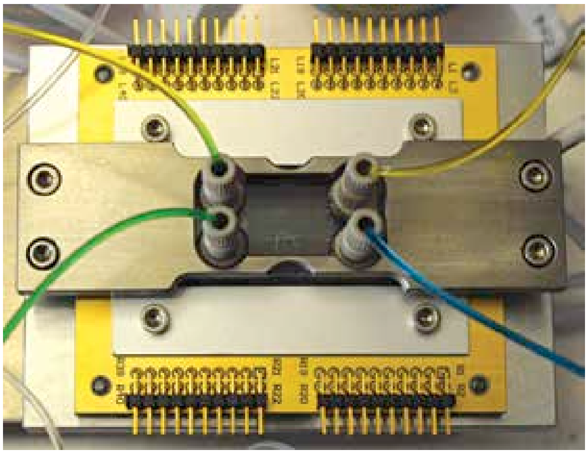
needed: (1) The power density of processors has to be reduced by improved architectures that minimize data motion and (2) the power density of electrochemical power delivery has to be massively improved. An example of the first effort is in the increased on-chip specialization to provide less power hungry compute systems that are based on efficiency improvements and educated compromises in computation and performance. Such devices can now also be used as servers for memory bound applications with better efficiency compared to current servers.

For the second effort experimental investigations of electrochemical conversion in single-channel redox flow systems have been triggered to explore approaches to overcome power density limitations. We believe that future computers which additionally could be built around this fluidic power delivery scheme would be less power-intensive due to their reduced communication power demand. The fluidic means of removing heat allows considerable increases in packaging density while the fluidic delivery of power with the same medium saves the space used by hardwired power delivery. The savings in space allow much denser architectures, sharply reduced energy needs and latency as communication paths shrink. Together with the improved power densities of redox flow batteries it should be possible to deliver the complete power demand of high-performance, high-density, processing architectures.

With the obsolescence of Moore's law, would it be right to say that something like Electronic Blood is more important now than it ever was?

Bruno: Until now, the number of transistors on a chip and hence computer performance has doubled every 18 months in accordance with Moore's Law. The downside is that their power consumption has also grown at the same pace. This trend of ever-increasing computer performance is expected to be pushed to the zeta scale in the next few decades. Such systems would be able to perform 1021 —a 1 followed by 21 zeros —computational operations per second, which is roughly 30,000 times more than today's best computer systems can do. However, based on current technologies, a supercomputer of this capacity would devour more than the entire global production of electricity.

IBM Research is investigating a number of promising research technology to address these challenges including quantum computing, neuromorphic computing, and nanotechnology. So these are all critical in addressing the end of Moore's Law, with each having different time horizons. The great strength of the "electronic blood" is that it is the final point of a new density roadmap that uses dense mobile and wearable packaging to get more compact systems and combines this with our research effort on 3D packaging to allow higher densification factors exceeding 1'000'000x. The roadmap starts immediately with the first proof points being the 1000x denser microserver. It builds on existing CMOS technology and is fully backward or legacy compatible in the software stack - capabilities that quantum computing and neuromorphic computing do not have. So we clearly can confirm that this density and electronic blood roadmap is much more important now than it ever was - in fact it replaces the obsolete Moore's law with a new "law" that is based on volumetric densification instead of areal densification.



Single-channel test chips used by IBM researchers

How far away do you think we are, from actually seeing this technology being put to use?

Bruno: We hope to see the first applications around 2030.

Can Electronic Blood also reduce the electricity consumption of a supercomputer, or does it act as a vehicle for transferring electricity?

Bruno: Yes, that is the goal. I address this answer in Q2.

This groundbreaking approach is modelled after the structure and power supply of the brain, where blood capillaries serve both to cool it and to supply it with energy. Compared to today's top computers, however, the human brain is roughly 10,000 times denser and 10,000 times more energy-efficient. The research team believes that their approach could reduce the size of a computer with a performance of 1 petaflop/s from the dimensions of a school classroom to that of an average PC, or in other words to a volume of about 10 liters.

Electronic Blood acts as a mediator of energy, not an energy source. However, by enabling to build computer systems with much higher density per unit volume, the resulting energy consumption of the computer for transferring power and communicating data can be reduced, resulting in a higher overall efficiency. Further, Electronic Blood could allow to reduce the number of voltage transformation steps in a datacenter.

Lastly, the technology obviously tries to replicate human blood, how important do you think it is, in the field of Artificial Intelligence?

Bruno: The concept is not tied to a specific field of computing or even device technology, as it primarily addresses the way a computer is powered and cooled.

Any applications requiring computers to process large amounts of structured and unstructured data can benefit from the technology we are developing in REPCOOL, because it will let users explore this data faster, using less energy.